



SCHOOL *of* DATA SCIENCE

Capstone Computing Resources

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DS 6015: Data Science Capstone
School of Data Science
University of Virginia

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Compute and Storage

- To complete your capstone, you'll need compute and storage
- Compute – e.g., running Python scripts to train models, run inference
- Storage – e.g., large amounts of data files
- This deck will outline the options and considerations

Agenda

- Sensitive Data
- Sponsor Provided Resources
- Computing on Your Own Machine
- UVA Research Computing Resources
- Example Scenarios

Sensitive Data

- The presence of sensitive data can complicate matters
- Sensitive data includes:
 - Personally Identifiable Information (PII)
 - Protected Health Information (PHI)
 - Company/business data
- Secure environments are a must for handling this data (e.g., Ivy)
- Users cannot handle this data on their laptops

Sensitive Data, contd.

- Not all capstones work with sensitive data
- In some cases, sponsors can remove this data, or replace with meaningful data values that are not sensitive (de-identify the data)

Sponsor Provided Resources

- In some cases, the sponsor might give you access to their platform
- This is generally signaled early by the sponsor
- Will require login credentials and possibly training
- The platform might be a cloud environment (e.g., AWS)
- You would use the suggested tools (e.g., cloud storage, database)

Sponsor Provided Resources, contd.

- Users must NOT move the data off the platform
- Users are generally given access to environment that is not production

Computing on Your Own Machine

- In some cases, you might be able to compute on your machine
- May be feasible when:
 - datasets are small
 - the data is not sensitive (e.g., no PII)

Computing on Your Own Machine, contd.

- You will want to have a shared location for data
- In some cases, the sponsor might have tools or suggestions
- e.g., Google Drive
- This can work for non-sensitive data

UVA Research Computing

- UVA has on-premises computing resources: High Performance Cluster (HPC)
- Your faculty mentor can request an allocation, storage for the team
- <https://www.rc.virginia.edu/>

Options:

> **Rivanna / Afton:** computing and storage

For non-sensitive data

> **Ivy** : Compute for sensitive data. <https://www.rc.virginia.edu/userinfo/ivy/>

High security VPN on machine (need training first)

Need OPSWAT software on machine

Can take 1-2 weeks onboard (watch for emails with instructions)

Example Compute Scenarios

- If the sponsor can provide computational resources, this is a great option
- If the sponsor cannot provide resources:
 - > For non-sensitive data
 - Large jobs: **Rivanna / Afton**
 - Small jobs: Local, Google Colab
 - > For sensitive data, use **Ivy**

Takeaways

- There are several options for compute / storage
- Start the compute / storage discussion early with your sponsor and mentor
- If you need UVA Research Computing (Rivanna / Afton or Ivy),
 - Email your faculty mentor and Capstone Director early
- Never store sensitive data on your machine or Rivanna